

# Portfolio Question - How many Eigenvalues to retain?

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Portfolio 03.1.P (v4.0.0)

## Portfolio Question 3.1.P - How many Eigenvalues to retain?

- ▶ How many dimensions should we retain in an Eigenvalue decomposition?
  - ▶ Is there an objective solution? Under what circumstances?

# How many PCs?

- ▶ We select only those PCs whose eigenvalues are “large enough”. Procedures include:
  - ▶ **scree plot**, looking for an elbow in the distribution
  - ▶ **EVs  $> 1$** , justified by random graph theory
  - ▶ **Tracey-Widom theory**, similarly
  - ▶ **Horn’s criterion**, based on simulating random matrices for the remaining matrix structure after  $K$  are chosen
  - ▶ **Velicer’s MAP criterion**, similar
- ▶ In practice any can be “wrong”, so common sense should be applied.
- ▶ Eigenvectors associated with Eigenvalues that are “too small” will contain some noise, even if they still contain a signal.

# How many PCs?

- ▶ In the data from the lecture, we have only 4 features. And we mean centred, which removes a degree of freedom. So the data should lie on a 3D subspace:

```
> round(testdata.eigen$values[1:4],8)
[1] 1859.25164 124.70157 16.04679 0.00000
> round(((testdata.svd$d)^2)/3,8)
[1] 1859.25164 124.70157 16.04679 0.00000
```

- ▶ Scaling by the number of features is not important for the SVD when computing the *proportion* of variance explained. Just square the singular values.

## Mathematical properties:

- ▶ If  $X$  is **positive definite**, its eigenvalues are  $> 0$ .
- ▶ If  $X$  is **positive semi-definite**, its eigenvalues are  $\geq 0$ .
- ▶ positive definite: If  $\forall v \neq 0$  then  $v \cdot xv > 0$
- ▶ positive semi-definite: If  $\forall v \neq 0$  then  $v \cdot xv \geq 0$
- ▶ The matrix  $A$  is **orthogonal** if  $A^T = A^{-1}$ . This is true iff all column vectors of  $A$  are orthonormal, and (equivalently) the row vectors are too. All eigenvalues of an orthogonal matrix are 1.
- ▶ If  $X$  is square and non-degenerate (distinct eigenvalues), its eigenvectors  $U$  form an **orthonormal basis**.

# Projections are idempotent

- ▶ Once you project a vector into a subspace, projecting it again does nothing. Such projections  $P$  are called **idempotent**:

$$P^K = P, \quad \forall K \geq 1$$

- ▶ Spectral projections have this property.

# Discussion

- ▶ How do we determine the elbow in a scree plot? Can we find examples with a discussion?
- ▶ Are any of the criterion “right”? When?
- ▶ Can we use cross-validation to obtain something more concrete? What would that look like?
  - ▶ Do we need a classification or similar supervised task? Is PCA useful for that?

# R commands for matrix operations

```
A %*% B # matrix multiplication  
t(A) # matrix transpose  
diag(A) # diagonal vector of A  
diag(x) # a diagonal matrix formed of the vector x  
det(A) # determinant  
sum(diag(A)) # trace  
solve(A) # matrix inverse  
eigen(A) # Eigenvalue decomposition  
svd(A) # Singular Value Decomposition
```